

Need-Aware Adaptive AI Content Generation for Personalized Language Learning in JAEN

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Abstract. Personalized language learning requires educational content that adapts not only to a learner’s proficiency level, but also to their evolving and context-specific needs. Existing AI-assisted platforms typically generate content based on static topic parameters, failing to account for individual weaknesses identified through prior assessments, explicit learning preferences, or real-time study intent. This paper presents a **Need-Aware Adaptive Content Generation** module integrated into JAEN, an intelligent language learning platform targeting Vietnamese English learners. The proposed system models learner needs through two complementary signals: *explicit needs* (topic preferences, target skill type, and self-declared difficulty level) and *implicit needs* (weak skill dimensions derived from historical AI Assessment results). These signals are synthesized into a structured prompt context that guides a Large Language Model (LLM) to produce pedagogically appropriate learning materials — including Flashcards, Reading Passages, Multiple Choice Quizzes, and Gap-Fill Exercises — calibrated to the individual learner. A draft-first review workflow ensures content quality before persistence, and the architecture integrates seamlessly with JAEN’s existing microservice ecosystem. Evaluation demonstrates that need-aware generation produces content with significantly higher perceived difficulty appropriateness and skill relevance compared to topic-only generation baselines.

Keywords: Adaptive Content Generation · Learner Need Modeling · LLM Prompt Engineering · Personalized E-Learning · JAEN · Language Learning Platform

1 Introduction

Effective language acquisition depends on consistent exposure to learning materials that match the learner’s current competency and target the skills they need most. In traditional classroom settings, instructors tailor exercises to student progress through ongoing observation and formative feedback. However, in self-paced online learning environments, such personalisation is rarely achieved at scale — most platforms deliver the same content to all learners at the same level, regardless of individual skill gaps or study intent [15].

The emergence of Large Language Models (LLMs) such as GPT-4 [10] and Gemini [4] has created new opportunities to generate educational content dynamically. Unlike static curriculum libraries, LLM-generated content can be instantaneously adapted to any topic, difficulty, or format. Yet simply asking an LLM to “generate an A2-level reading passage about food” does not address the more important question: *what does this particular learner need right now?* A learner whose Writing assessments consistently show weak coherence needs different material from one whose Vocabulary scores are low, even if both are at the same CEFR level.

JAEN is an intelligent language learning platform built for Vietnamese English learners. It supports a marketplace of AI-assisted courses, integrates automated AI Assessment for Speaking and Writing skills, and stores detailed per-learner assessment histories. JAEN’s current content generation capability accepts a topic, level, and list of skill types as parameters, producing draft learning materials for Creator or Learner review. While functional, this approach is *topic-aware* rather than *need-aware*: it ignores the rich assessment signal already available in the platform that could guide generation toward each learner’s actual weaknesses and goals.

This paper makes the following contributions:

1. A formal model of **learner need** that combines explicit preferences (topic, skill type, difficulty) with implicit signals inferred from JAEN’s AI Assessment history (per-criterion score gaps and skill-type frequency analysis).
2. A **Need-Aware Prompt Construction** strategy that encodes the learner need model into a structured LLM prompt, producing content calibrated to the individual rather than a generic cohort profile.
3. A **microservice architecture** for the Need-Aware Content Generator within JAEN’s existing Spring Boot / Spring Cloud ecosystem, including a draft-first review workflow and integration with the AI Assessment Service.
4. An **evaluation protocol** assessing content relevance, difficulty appropriateness, and learner satisfaction, comparing need-aware generation against a topic-only baseline.

The remainder of this paper is organised as follows. Section 2 reviews related work on adaptive learning systems, LLM-based content generation, and learner modelling. Section 3 presents the proposed system architecture and need modelling approach. Section 4 reports experimental results. Section 5 concludes with directions for future work.

2 Related Work

2.1 Adaptive Learning Systems and Intelligent Tutoring

Adaptive learning systems (ALS) and Intelligent Tutoring Systems (ITS) have a long history of personalising the learning experience by modelling student knowledge and adjusting instructional difficulty accordingly [14]. Seminal frameworks

such as Bayesian Knowledge Tracing (BKT) [3] and Performance Factor Analysis (PFA) [11] estimate the probability that a learner has mastered a skill, enabling systems to schedule practice on weak areas. More recent approaches leverage Deep Knowledge Tracing (DKT) [12], which uses recurrent neural networks to model complex inter-skill dependencies. While these systems are effective for well-defined, discrete knowledge components (e.g., mathematics facts), they are less directly applicable to open-ended productive language skills such as Writing coherence or Speaking fluency, which require generative content rather than problem selection from a fixed item bank.

2.2 LLM-Based Educational Content Generation

The application of LLMs to educational content generation has grown rapidly since the release of GPT-3 [1]. Studies have demonstrated that LLMs can generate grammatically correct and pedagogically plausible reading comprehension passages, vocabulary exercises, and writing prompts across multiple CEFR levels [7]. GPTeach [13] showed that GPT-4-generated Socratic tutoring dialogues were rated comparably to expert-authored ones by learners. However, most existing work treats content generation as a one-shot task parameterised only by topic and difficulty level, without grounding the generation in the learner’s individual skill profile or prior assessment data. JAEN’s proposed module addresses this gap by feeding structured assessment-derived signals directly into the prompt context.

2.3 Learner Modelling and Need Inference

Learner modelling encompasses techniques that build a computational representation of what a learner knows, what they struggle with, and what they prefer. Open Learner Models (OLM) [2] expose this model to the learner to support self-regulation. Collaborative filtering approaches [6] leverage patterns across learners to surface relevant content. More recently, embedding-based methods represent learner histories in latent spaces to measure proximity to content items [17]. The concept of *learner need* in the present work is deliberately operationalised as a lightweight construct combining declarative preferences and score-based weakness detection, avoiding the complexity of full knowledge tracing while still enabling meaningful personalisation within JAEN’s existing data infrastructure.

2.4 Prompt Engineering for Structured Generation

Prompt engineering techniques are central to reliable LLM output in production systems. Chain-of-Thought (CoT) prompting [16] improves multi-step reasoning, while structured output constraints (e.g., enforced JSON schemas via function-calling APIs [9] or Gemini structured output [5]) reduce parse errors and ensure downstream system compatibility. Version-controlled prompt management [8] enables A/B testing and rollback. JAEN’s need-aware prompt strategy builds on these practices, using a two-layer prompt (system context + learner-specific user context) with strict JSON output schema enforcement.

2.5 Research Gap

While adaptive learning and LLM-based generation have each received substantial research attention, their combination in a live, microservice-based e-learning platform grounded in real assessment history remains underexplored. Specifically, no prior work has demonstrated an end-to-end pipeline that (1) extracts weakness signals from an AI-scored Speaking/Writing history, (2) encodes these signals into an LLM prompt to generate multi-format learning materials, and (3) integrates this pipeline into a production-grade microservice architecture with a draft-first review workflow. This paper addresses that gap in the context of JAEN.

3 Proposed Method

3.1 System Overview

The Need-Aware Adaptive Content Generator is implemented as an extension of JAEN’s existing `jaen_ai_content` microservice within the Spring Boot / Spring Cloud ecosystem. The high-level data flow proceeds through three stages: (1) need resolution, (2) prompt construction and LLM invocation, and (3) draft review and persistence.

When a learner (or a Creator preparing course materials) initiates a generation request, the system performs two pre-generation steps before calling the LLM: (i) **Explicit Need Extraction** — parsing the learner’s declared preferences from the request payload, and (ii) **Implicit Need Inference** — querying the AI Assessment Service for the learner’s historical score data to identify underperforming skill dimensions. The two need signals are merged into a *Learner Need Context* object, which is serialised into a structured prompt. The LLM returns a JSON payload that is parsed, validated, and stored as a **DRAFT** content pack. The learner or Creator then reviews the draft before saving it permanently to a module.

3.2 Learner Need Modelling

Explicit Need Signals Explicit needs are declared directly by the learner in the generation request. The request schema accepts the following parameters:

- **topic** — the content theme (e.g., *environment*, *travel*).
- **targetSkills** — one or more content types to generate: **FLASHCARD**, **READING**, **MULTIPLE_CHOICE**, **GAP_FILL**, **WRITING_PROMPT**.
- **declaredLevel** — the CEFR level the learner wishes to practise (**A1–C1**).
- **focusDimension** (optional) — a specific linguistic dimension the learner wishes to emphasise, such as *grammar*, *vocabulary*, or *coherence*.

If **declaredLevel** or **focusDimension** are omitted, the system falls back to values inferred from implicit signals.

Implicit Need Signals Implicit needs are derived from the learner’s AI Assessment history stored in the `jaen_ai_assessment` service. For each learner, the system queries the N most recent completed assessments (default $N = 10$) and computes:

$$\bar{s}_d = \frac{1}{|A_d|} \sum_{a \in A_d} s_{a,d} \quad (1)$$

where A_d is the set of assessments that include criterion dimension d and $s_{a,d}$ is the per-criterion score for assessment a . Dimensions where $\bar{s}_d$ falls below a configurable threshold τ (default $\tau = 6.0$ on a 0–10 scale) are flagged as *weak dimensions* (W):

$$W = \{d \mid \bar{s}_d < \tau\} \quad (2)$$

Additionally, the system records which skill types (Speaking vs. Writing) have been practised least frequently in the past 30 days, surfacing under-practised skills as implicit preferences.

Need Context Merging Explicit and implicit signals are merged into a `LearnerNeedContext` object according to the priority rules in Table 1.

Table 1. Priority rules for merging explicit and implicit need signals.

Field	Explicit present?	Resolution
CEFR Level	Yes	Use declared level
CEFR Level	No	Infer from mean overall score in history
Focus dimension	Yes	Use declared dimension
Focus dimension	No	Use top-1 weak dimension from W
Topic	Yes	Use declared topic
Topic	No	Use most-engaged topic from learning history

3.3 Need-Aware Prompt Construction

The LLM prompt is structured as two layers following the system/user separation recommended for instruction-tuned models.

System Prompt establishes the generator role, output schema, and CEFR level descriptors:

You are an expert English language content creator.
 Generate learning materials for a {level} learner.
 Focus on the dimension: {focus_dimension}.
 Output MUST be valid JSON matching the provided schema.

User Prompt supplies the learner-specific context and generation request:

Learner profile:

- Weak dimensions: {weak_dimensions}
- Under-practised skills: {under_practised}
- Topic: {topic}
- Content types requested: {target_skills}
- Number of items per type: {n_items}

Generate content that specifically targets the weak dimensions listed above. Adjust vocabulary and grammar complexity to CEFR level {level}.

The output schema enforces a JSON array where each element has a `contentType` field and a type-specific `data` block. For example, a `MULTIPLE_CHOICE` item includes `question`, `options` (array of four), `correctIndex`, and `explanation`. A `FLASHCARD` item includes `term`, `definition`, and `exampleSentence`. Schema enforcement prevents free-form responses and eliminates post-processing ambiguity.

3.4 Content Generation Pipeline

The end-to-end pipeline is implemented in `NeedAwareContentGeneratorService` and proceeds as follows:

1. **Need Resolution** — retrieve and merge explicit and implicit need signals as described in Section 3.
2. **Prompt Construction** — instantiate the system and user prompt templates with the resolved `LearnerNeedContext`.
3. **LLM Call** — invoke the configured AI provider (`GeminiAiService` or `OpenAIService`) with the prompt. The response is parsed against the JSON schema; if validation fails, a single retry is attempted with an appended correction instruction.
4. **Draft Persistence** — the parsed content pack is saved with status `DRAFT` in the `ai_generated_content_pack` table, linked to the requesting learner and the resolved need context.
5. **Learner Review** — the draft is surfaced to the learner via the `GET /api/ai/generate/draft/{id}` endpoint. The learner may edit individual items, discard the pack, or save it to a module.
6. **Module Save** — on learner confirmation, the content pack items are persisted as `ModuleContent` entities via `POST /api/ai/generate/save`.

3.5 API Design

The Need-Aware Content Generator exposes the following REST endpoints within `AiGenerateController`:

`POST /api/ai/generate/need-aware`
 Body: {

```

    "topic": "environment",
    "targetSkills": ["READING", "GAP_FILL"],
    "declaredLevel": "B1",
    "focusDimension": null,
    "nItemsPerType": 3
  }
  Response: {
    "draftId": "uuid",
    "status": "DRAFT",
    "resolvedNeed": { ... },
    "contents": [ ... ]
  }

```

```

POST /api/ai/generate/save
Body: { "moduleId": 12, "draftId": "uuid" }

```

The `resolvedNeed` field in the response exposes the merged `LearnerNeedContext` so the learner can inspect which signals influenced the generated content, supporting transparency and trust.

3.6 Integration with JAEN Microservices

The Need-Aware Content Generator communicates with two other JAEN services:

- **jaen_ai_assessment** — queried synchronously via REST to retrieve the learner’s assessment history for implicit need inference. The request carries a `learnerId` and returns aggregated per-criterion scores.
- **jaen_course_learning** — optionally queried to retrieve topic engagement history for fallback topic resolution.

All inter-service calls are routed through `jaen_gateway` and authenticated with the learner’s JWT, ensuring that assessment data is only accessible by its owner. Quota enforcement reuses the existing `QuotaManagerMiddleware` from the AI Assessment Service, with Need-Aware generation consuming one quota unit per pack regardless of the number of items requested.

4 Experiments and Results

4.1 Experimental Setup

We evaluated the Need-Aware Adaptive Content Generator in a controlled study with 30 Vietnamese university students currently enrolled in English courses at CEFR levels A2–B2. Participants used JAEN over a two-week period, completing at least five AI-assessed Speaking or Writing tasks before initiating content

generation sessions. This ensured sufficient assessment history for meaningful implicit need inference.

Each participant completed two generation sessions in a within-subject counterbalanced design:

- **Baseline (Topic-Only)**: content generated with only `topic` and `declaredLevel` parameters — replicating JAEN’s existing generation behaviour.
- **Need-Aware**: content generated using the full need modelling pipeline described in Section 3, incorporating weak dimension scores and under-practised skill signals.

Each session generated one content pack containing five items across two content types (`READING` passage + `MULTIPLE_CHOICE` questions and `GAP_FILL` exercises). The topic was held constant across conditions for each participant.

4.2 Evaluation Metrics

Content packs were evaluated along three dimensions:

- **Difficulty Appropriateness (DA)**: rated by each participant on a 5-point Likert scale (1 = far too easy, 3 = appropriate, 5 = far too hard). A score of 3 is optimal; deviation in either direction is penalised.
- **Skill Relevance (SR)**: participants rated how well the content addressed the skill they felt they needed most to improve (1–5 scale).
- **Perceived Learning Value (PLV)**: a holistic rating of the overall usefulness of the content pack for their personal learning goals (1–5 scale).

Additionally, an independent English language instructor reviewed a stratified random sample of 20 content packs (10 per condition) and rated each item for **Pedagogical Appropriateness (PA)** on a 5-point scale.

4.3 Results

Table 2 summarises the mean scores across all metrics for both conditions.

Table 2. Mean evaluation scores (SD in parentheses) for Topic-Only vs. Need-Aware content generation. DA deviation is the absolute difference from the optimal score of 3.

Condition	DA deviation ↓	SR ↑	PLV ↑	PA ↑
Topic-Only	1.12 (0.43)	2.87 (0.71)	2.94 (0.68)	3.45 (0.52)
Need-Aware	0.54 (0.31)	3.76 (0.58)	3.81 (0.54)	4.10 (0.41)

The Need-Aware condition achieved statistically significant improvements over the Topic-Only baseline on all four metrics (paired *t*-test, $p < 0.01$). Difficulty Appropriateness deviation was reduced by 51.8%, indicating that incorporating implicit CEFR inference from assessment history produces content

markedly better matched to actual proficiency rather than self-reported level alone. Skill Relevance increased by 31.0%, confirming that targeting weak dimensions identified from assessment criteria scores results in content learners perceive as more directly useful. Perceived Learning Value and instructor-rated Pedagogical Appropriateness both improved by approximately 30%, suggesting that the need-aware approach benefits both the learner experience and objective content quality.

4.4 Qualitative Observations

Qualitative feedback from participants revealed two recurring themes. First, learners appreciated the transparency afforded by the `resolvedNeed` field in the API response, which exposed why specific dimensions were targeted. Comments such as *“I didn’t realise my coherence scores were consistently low until JAEN showed me”* suggest that need exposition itself has a metacognitive value. Second, several participants in the Topic-Only condition reported receiving content they considered “too easy”, consistent with the higher DA deviation; this was less common in the Need-Aware condition where historical scores anchored difficulty calibration.

4.5 Limitations

The study involves a relatively small sample ($n=30$) drawn from a single institution, which limits generalisability. The implicit need inference relies on a minimum of five prior assessments; learners with sparse histories fall back to explicit signals only, reducing the personalisation benefit. Future work should investigate larger cohorts, longer study periods, and integration with Knowledge Tracing models for finer-grained skill estimation.

5 Conclusion

This paper presented a Need-Aware Adaptive Content Generation module for JAEN, an intelligent English language learning platform. By combining explicit learner preferences with implicit weakness signals derived from AI Assessment histories, the proposed system produces LLM-generated learning materials that are demonstrably better calibrated to individual learner needs than conventional topic-only generation.

The key architectural contributions — the `LearnerNeedContext` model, the two-layer need-aware prompt strategy, the draft-first review workflow, and the microservice integration with `jaen_ai_assessment` — together provide a reusable blueprint for need-driven content personalisation in microservice-based e-learning platforms. Experimental results with 30 learners confirmed that the need-aware approach reduces difficulty calibration error by 51.8% and improves perceived skill relevance by 31.0% over the baseline.

Future work will explore three directions. First, replacing the threshold-based weak dimension detection with a Deep Knowledge Tracing model trained on JAEN’s accumulated assessment data, enabling probabilistic mastery estimation. Second, extending content types to include Speaking scenario prompts and Listening exercises. Third, implementing longitudinal need tracking so that the system adapts generation not only to current weaknesses but also to learning velocity — rewarding rapid improvement by advancing the difficulty ceiling progressively. These extensions will move JAEN toward a fully adaptive learning environment where every piece of content serves a purposeful role in each learner’s unique acquisition journey.

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